

Methodology of Outside Field Dosimetry

Acuros as algorithm and Halcyon as a LINAC



Souvik Das

Resident Medical Physicist
Tata Memorial Hospital, Mumbai

Table of Contents

1 Project topic

Out of Field Dosimetry in Halcyon using Acuros as algorithm.

Equipment required:

- Halcyon LINAC
- Eclipse Treatment Planning System
- ATOM anthropomorphic male reference phantom (CIRS, Inc., Norfolk, VA)
- LD

Which Site to be investigated?

I have to make a dataset which type of patient are lived long after the radiation therapy.

ATOM anthropomorphic male reference phantom

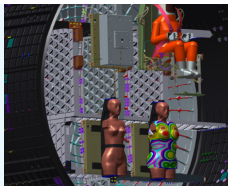
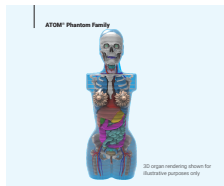
Description	Model	Available Dosimetry Configurations	Height, cm	Weight, kg	Thorax Dimensions, cm	Bone Physical Density, cc	Bone Electron Density, 1/cc
Adult Male	701	B, C, D & G	173	73	23 cm $\times$ 32 cm	1.60	5.030×10^{23}

Configuration key:

- B Configuration = 3×3 cm Grid - 5 mm Holes
- C Configuration = 1.5×1.5 cm Grid - 5 mm Holes
- D Configuration = Organ Grid - 5 mm Holes
- G Configuration = Organ Grid - 14 mm Holes

ATOM anthropomorphic male reference phantom

Manufacturer slide



Enabling Safer Human Space Travel

NASA's Artemis I mission to the Moon in 2022 featured two custom designed female ATOM Phantoms. "Helga" and "Zohar" flew as proxies for female astronauts to help determine risk to radiation sensitive organs, and ensure more inclusive possibilities for the future of human space travel.

Learn more: sunnuclear.com/news

Images source: The European Space Agency and DLR



Plug Configurations

Expanding Functionality

Regardless of the grid configuration or hole diameter, all holes in an ATOM phantom come profiled with an appropriate, tissue equivalent solid plug. These solid plugs can be easily removed and replaced with a dosimeter holder plug for available dosimetry configurations. These special plugs are designed to house a variety of dosimeters, including TLD chips, TLD rods, TLD glass rods, MOSFET detectors, and OSL holders. This pairing of dosimeter and plug protects the dosimeter and allows it to be placed in any hole in the phantom, with minimal air interface between the device and surrounding tissue.

All plugs are sold as kits that contain a variety of tissue types to ensure there is an equivalent plug for any dosimetry configuration and anatomical location

Thermoluminescence Dosimetry (TLD)

Sectional and drilled ATOM phantoms readily accept TLDs using a 5 mm plug. These plugs are manufactured to hold TLD chips, rods, bars, cubes, and discs in place during the dosimetry study. In applications where varying the depth of the dosimeter within the slab is important, such as in organ dosimetry, these rods can be easily modified by the user to custom depths within the slab.

ATOM® Phantom Family



Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

This specially machined plug can hold MOSFET dosimeters. The plug is manufactured to fit 5 mm diameter holes in an ATOM phantom. A recessed section at the center of the plug accepts the MOSFET sensitive volume and a radius on the end of the plug prevents harsh bends in the cable to prevent damage to the cable when the phantom is assembled.

OSL Dosimetry

OSL holder plugs are designed to fit standard OSL dosimeters and require 14 mm diameter hole configurations to accommodate the device.

Ion Chamber Dosimetry

ATOM phantoms can be custom drilled to accommodate various ion chambers and diodes. The Chamber cavity is drilled in the lateral direction along the midplane of the section. Each chamber cavity is supplied with a soft tissue equivalent plug. Upon order, customers should specify the desired anatomical location of the sensitive volume for the ion chamber as the location of the chamber will affect some holes in the grid configuration.

Description	Part #	Plug Size	Kit Contents
TLD Chip	Part numbers available upon request	5 mm Ø x 25 mm long	5 sets of 10 plugs (soft tissue, lung, brain, adult bone and pediatric bone plugs)
TLD / Glass Rod		5 mm Ø x 25 mm long (Ø 1.5mm Ø x 12 mm deep)	
MOSFET		5 mm Ø x 25 mm long	
OSL		14 mm Ø x 25 mm long	5 sets of 10 plugs (soft tissue, lung, brain, adult bone and pediatric bone plugs)
Ion Chambers		By quotation only, contact customer service for a quote	

Which Luminescence Dosimeter(LD)?

See